Linux Distro Command Cheat Sheet

Side-by-side equivalents for Debian/Ubuntu, Fedora/RHEL, Arch, openSUSE and Alpine. Replace `pkg`, `svc` and `user` with your own values. Most commands need `sudo` (Alpine: `doas` or root).

1. The distro families at a glance

	Debian / Ubuntu	Fedora / RHEL	Arch	openSUSE	Alpine
Based on	Debian	Red Hat	Independent	SUSE	Independent (musl, BusyBox)
Package format	.deb	.rpm	.pkg.tar.zst	.rpm	.apk
High-level tool	apt	dnf (older: yum)	pacman	zypper	apk
Low-level tool	dpkg	rpm	pacman	rpm	apk
Release model	Ubuntu: fixed, LTS every 2 yrs	Fedora: ~6 months RHEL: fixed, long-term	Rolling	Leap: fixed Tumbleweed: rolling	Fixed, ~6 months
Init system	systemd	systemd	systemd	systemd	OpenRC
Default firewall	ufw (Ubuntu, off by default)	firewalld	none installed	firewalld	none installed (use nftables/iptables)
Security module	AppArmor	SELinux	none by default	AppArmor (newer: SELinux)	none by default
Admin group	sudo	wheel	wheel	wheel	wheel
Extra software	snap, PPAs	Flatpak, COPR, EPEL	AUR (via yay/paru)	OBS, Packman	community repo, testing repo

2. Package management

Task	Debian / Ubuntu	Fedora / RHEL	Arch	openSUSE	Alpine
Refresh package index	`apt update`	`dnf makecache`	`pacman -Sy` `(never run alone, see note)`	`zypper refresh`	`apk update`
Upgrade everything	`apt upgrade` `apt full-upgrade`	`dnf upgrade`	`pacman -Syu`	`zypper update` `Tumbleweed: zypper dup`	`apk upgrade`
Install a package	`apt install pkg`	`dnf install pkg`	`pacman -S pkg`	`zypper install pkg`	`apk add pkg`
Remove a package	`apt remove pkg`	`dnf remove pkg`	`pacman -R pkg`	`zypper remove pkg`	`apk del pkg`
Remove + config + unused deps	`apt purge pkg` `apt autoremove`	`dnf remove pkg` `dnf autoremove`	`pacman -Rns pkg`	`zypper remove --clean-deps pkg`	`apk del pkg`
Search repos	`apt search term`	`dnf search term`	`pacman -Ss term`	`zypper search term`	`apk search term`
Show package info	`apt show pkg`	`dnf info pkg`	`pacman -Si pkg (repo)` `pacman -Qi pkg (installed)`	`zypper info pkg`	`apk info pkg`
List installed packages	`apt list --installed` `dpkg -l`	`dnf list installed` `rpm -qa`	`pacman -Q`	`zypper search -i` `rpm -qa`	`apk info`
Which package owns a file	`dpkg -S /path/file`	`rpm -qf /path/file`	`pacman -Qo /path/file`	`rpm -qf /path/file`	`apk info --who-owns /path/file`
List files in a package	`dpkg -L pkg`	`rpm -ql pkg`	`pacman -Ql pkg`	`rpm -ql pkg`	`apk info -L pkg`
Which package provides a file / command	`apt-file search name` `(install apt-file first)`	`dnf provides */name`	`pacman -F name` `(run pacman -Fy first)`	`zypper search --provides name`	`apk search cmd:name`
Install a local package file	`apt install ./file.deb` `dpkg -i file.deb`	`dnf install ./file.rpm`	`pacman -U file.pkg.tar.zst`	`zypper install ./file.rpm`	`apk add --allow-untrusted file.apk`
Clean package cache	`apt clean`	`dnf clean all`	`pacman -Sc` `(or paccache -r)`	`zypper clean`	`apk cache clean`
Hold / pin a version	`apt-mark hold pkg`	`dnf versionlock add pkg` `(needs the versionlock plugin)`	`add to IgnorePkg in /etc/pacman.conf`	`zypper addlock pkg`	`apk add pkg=version`
Find orphaned packages	`apt autoremove --dry-run`	`dnf autoremove --assumeno`	`pacman -Qdtq`	`zypper packages --orphaned`	`n/a`
Add a repository	`add-apt-repository ppa:x/y` `or edit /etc/apt/sources.list.d/`	`dnf copr enable user/proj` `dnf config-manager addrepo --from-repofile=URL`	`edit /etc/pacman.conf` `(AUR: git clone + makepkg -si, or yay)`	`zypper addrepo URL name`	`edit /etc/apk/repositories`
Show repo list	`cat /etc/apt/sources.list` `ls /etc/apt/sources.list.d/`	`dnf repolist`	`cat /etc/pacman.conf`	`zypper repos`	`cat /etc/apk/repositories`
View package history	`less /var/log/apt/history.log`	`dnf history`	`less /var/log/pacman.log`	`less /var/log/zypp/history`	`n/a`

Arch note: it does not support partial upgrades. Always run `pacman -Syu` as one command rather than `-Sy` followed by installing a package. On Debian/Ubuntu, older guides use `apt-get` and `apt-cache`; they still work and behave the same in scripts. RHEL 7 and CentOS 7 use `yum` instead of `dnf`.

3. Services and system management

Task	Debian / Ubuntu	Fedora / RHEL	Arch	openSUSE	Alpine
Start / stop a service	`systemctl start svc` `systemctl stop svc`	same	same	same	`rc-service svc start` `rc-service svc stop`
Restart / reload	`systemctl restart svc` `systemctl reload svc`	same	same	same	`rc-service svc restart`
Enable / disable at boot	`systemctl enable svc` `systemctl disable svc`	same	same	same	`rc-update add svc default` `rc-update del svc default`
Check status	`systemctl status svc`	same	same	same	`rc-service svc status`
List running services	`systemctl list-units --type=service`	same	same	same	`rc-status`
View logs	`journalctl -u svc` `journalctl -b`	same	same	same	`less /var/log/messages` (syslog, not journald)
Reboot / shutdown	`reboot` `shutdown -h now`	same	same	same	`reboot` `poweroff`
Rebuild initramfs	`update-initramfs -u`	`dracut --force`	`mkinitcpio -P`	`dracut -f` (or `mkinitrd`)	`mkinitfs`
Update GRUB config	`update-grub`	`grub2-mkconfig -o /boot/grub2/grub.cfg`	`grub-mkconfig -o /boot/grub/grub.cfg`	`grub2-mkconfig -o /boot/grub2/grub.cfg`	varies (syslinux or GRUB)
Set hostname	`hostnamectl set-hostname name`	same	same	same	`echo name > /etc/hostname` `hostname -F /etc/hostname`
Set timezone	`timedatectl set-timezone Zone/City`	same	same	same	`setup-timezone` (or link `/etc/localtime`)
Set locale	`dpkg-reconfigure locales` or `localectl set-locale`	`localectl set-locale LANG=en_US.UTF-8`	edit `/etc/locale.gen`, run `locale-gen`, then `localectl`	`localectl set-locale`	set `LANG` in `/etc/profile.d/`

"same" means the command in the first column works unchanged.

4. Users, groups and networking

Task	Debian / Ubuntu	Fedora / RHEL	Arch	openSUSE	Alpine

Add user with home dir	`adduser user`
	`(friendly script)`
	`useradd -m user`
		`useradd -m user`	`useradd -m user`	`useradd -m user`	`adduser user`

Give a user admin rights	`usermod -aG sudo user`	`usermod -aG wheel user`	`usermod -aG wheel user` `(enable wheel in sudoers)`	`usermod -aG wheel user`	`adduser user wheel` `(install sudo or doas)`
Edit sudoers safely	`visudo`	`visudo`	`visudo`	`visudo`	`visudo` `or edit /etc/doas.d/`

Network config	`netplan (Ubuntu):`
	`/etc/netplan/*.yaml`
	`then: netplan apply`
	`Debian:`
	`/etc/network/interfaces`
		`NetworkManager:`
		`nmcli, nmtui`
			`systemd-networkd`
			`or NetworkManager`
				`NetworkManager (desktop)`
				`wicked (server, older)`
					`/etc/network/interfaces`
					`setup-interfaces`

Show IP addresses	`ip a`	`ip a`	`ip a`	`ip a`	`ip a`

Manage firewall	`ufw allow 22/tcp`
	`ufw enable`
		`firewall-cmd`
		`--add-service=ssh`
		`--permanent`
		`firewall-cmd --reload`
			`nft / iptables`
			`(or install ufw)`
				`firewall-cmd (same as Fedora)`	`nft / iptables`

SELinux / AppArmor status	`aa-status`	`getenforce` `sestatus`	`n/a by default`	`aa-status` `(or sestatus)`	`n/a by default`

5. Package names that differ between distros

Task	Debian / Ubuntu	Fedora / RHEL	Arch	openSUSE	Alpine
Compiler + build tools	`build-essential`	`dnf group install` `development-tools`	`base-devel`	`zypper install -t pattern` `devel_basis`	`build-base`
Kernel headers	`linux-headers-$(uname -r)`	`kernel-devel`	`linux-headers`	`kernel-devel`	`linux-lts-dev`
Library dev files (example: openssl)	`libssl-dev`	`openssl-devel`	`openssl` `(headers included)`	`libopenssl-devel`	`openssl-dev`
Apache web server	`apache2`	`httpd`	`apache`	`apache2`	`apache2`
SSH server	`openssh-server`	`openssh-server`	`openssh`	`openssh`	`openssh-server`
Python pip	`python3-pip`	`python3-pip`	`python-pip`	`python3-pip`	`py3-pip`
Vim (full version)	`vim`	`vim-enhanced`	`vim`	`vim`	`vim`
Cron	`cron`	`cronie`	`cronie`	`cron`	`busybox cron or dcron`
DNS tools (dig, nslookup)	`dnsutils / bind9-dnsutils`	`bind-utils`	`bind`	`bind-utils`	`bind-tools`
Netstat / ifconfig	`net-tools`	`net-tools`	`net-tools`	`net-tools`	`net-tools`
7-Zip	`p7zip-full`	`p7zip` `(EPEL on RHEL)`	`p7zip`	`p7zip-full`	`7zip`
Fonts (Microsoft-style)	`ttf-mscorefonts-installer`	`manual install`	`ttf-liberation`	`liberation-fonts`	`font-liberation`

Package names change between releases, so confirm with the search command from section 2 if an install fails. Library "dev" packages are the most common cause of confusion: Debian uses `-dev`, Fedora/openSUSE use `-devel`, and Arch ships headers inside the main package.

6. Commands that are the same everywhere

```
ls cd cp mv rm mkdir cat less grep find sed awk tar curl wget                    ip a show network addresses
chmod chown ps top kill df du free ssh scp rsync git nano vim systemctl*         ss -tulpn listening ports
                                                                                 uname -r kernel version
                                                                                 cat /etc/os-release identify the distro
                                                                                 lsblk / lsusb / lspci hardware
```

**systemctl is not on Alpine (it uses OpenRC). Alpine also uses BusyBox versions of many tools, so some flags differ from the GNU versions.*

7. Quick migration tips: Ubuntu to Arch

- Learn one habit first: `sudo apt update && sudo apt upgrade` becomes `sudo pacman -Syu`. It is a single step on Arch.
- Services are not started automatically after install on Arch. Run `systemctl enable --now svc` yourself.
- For software not in the official repos, use the AUR with a helper such as `yay -S pkg`. Read the PKGBUILD before installing.
- Arch config lives mostly in plain files (`/etc/pacman.conf`, `/etc/mkinitcpio.conf`) with no Ubuntu-style wrappers like netplan or ufw by default.
- Check `/var/log/pacman.log` and the Arch Wiki first when something breaks; the wiki is the best Linux documentation available and applies to most distros.
- After a kernel update on Arch, reboot before loading new kernel modules; the old modules are removed.